Date: Roll No.

III SEMESTER B.Tech (IT, ITNS)

END-SEMESTER EXAMINATION 2025

Course Code- INECC305/ITECC305/INECC06/ITECC06

Course Title-Probability & Stoch. Proc.

Time: 3:00 Hrs Max Marks: 50

Attempt all questions. Missing data / information (If any) may be suitably assumed & mentioned in the answer.

Q. No.		Questions	Marks	CO
Q 1		**Attempt any 2 out of 3 parts**		
	a	Events A and B are independent. Determine which of the following are true and which are false. Explain the reasons. i. $P(A\|B) = P(A)$ ii. $P(A \cup B\|C) = P(A\|C) + P(B\|C)$ iii. $P(A) = 0$, or $P(B) = 0$, or both. iv. $\frac{P(A\|B)}{P(B)} = \frac{P(B\|A)}{p(A)}$ v. $P(AB) = P(A)P(B)$	05	CO1
	b	Prove the following identity $P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(AB) - P(AC) - P(BC) + P(ABC)$	05	
	c	In the experiment of throwing two fair dice, let ***A*** be the event that the first die is odd, ***B*** be the event that the second die is odd, and ***C*** be the event that the sum is odd. Show that the events ***A***, ***B***, and ***C*** are pair wise independent, but ***A***, ***B***, and ***C*** are not independent.	05	
Q 2		**Attempt any 2 out of 3 parts**		
	a	Consider the function given by $F(x) = \begin{cases} 0, & x < 0 \\ x + \frac{1}{2}, & 0 \le x \le 1/2 \\ 1, & x \ge 1/2 \end{cases}$ i Sketch F(x) and show that F(x) has the properties of a cdf. ii If X is the random variable whose cdf is given by F(x), find 1. $P(X \le 1/4)$, 2. $P(0 < X \le 1/4)$, 3. $P(0 \le X \le 1/4)$. iii Specify the type of X.	05	CO2
	b	Given a random variable X that is uniformly distributed on [0,1]. A Random variable Y is defined as, $Y = \left(X - \frac{1}{2}\right)^2$. i Compute Prob $(Y > 1/8)$. ii Determine the mean and variance of Y.	05	
	c	The random variable ***X*** has PMF $P_X(x) = \begin{cases} c/x & x = 2, 4, 8 \\ 0 & otherwise \end{cases}$ i What is the value of constant ***c***? ii What is $P[X = 4]$? iii What is $P[X < 4]$? iv What is $P[3 \le X \le 9]$? v Compute mean of X.	05	
Q 3		**Attempt any 2 out of 3 parts**		
	a	Two random variables X and Y have the joint density function $f_{X,}(x,y) = \begin{cases} k\ (x + y/2)^2 & 0 < x < 2 \text{ and } 0 < y < 3 \\ 0 & elsewhere \end{cases}$ i Find k. ii Find the mean, variance, covariance of X,Y. iii Compute the probability of $\{X < Y\}$.	05	CO 3
	b	Given that X, Y are Gaussian Random Variables with m_X=1, m_Y=2, σ_X^2=1, σ_Y^2=4 & $E[XY] = 3$. Define $W = X + Y$ and $V = 2X - 3Y$. Find the PDF of W. Are W, V uncorrelated?	05	
	c	***X*** and ***Y*** are independent, identically distributed (i.i.d.) random variables with common p.d.f. $f_X(x) = e^{-x}U(x) \qquad f_Y(y) = e^{-y}U(y)$ Find the p.d.f. of the following random variables (1)$Z = X - Y$ (2)$R = XY$	05	
Q 4		**Attempt any 2 out of 3 parts**		
	a	Given a random process $X(t)$ defined as $X(t) = Ae^{-Bt}, \quad t > 0$ Where A and B are independent uniform random variables on $[-1,1]$ and $[0,1]$ respectively. i Find the first order density for the process. ii Compute $E[X(t)]$ the mean of the process. iii Determine $R_{XX}(t,u)$, the autocorrelation function for the process. iv Is $X(t)$ wide sense stationary?	05	CO 4

	b	Let X(t) be a continuous-time Gaussian WSS process with mean m_X=1 and $$R_X(\tau) = \begin{cases} 3 - \lvert\tau\rvert & -2 \le \tau \le 2 \\ 1 & else\ where \end{cases}$$ 1. Find the joint PDF of X(3), X(-2) 2. Find $E[((X(1) + X(2) + X(3))^2]$.	05	
		Describe Random Walk Process in detail. Derive the expressions of the second order joint PMF, mean and autocorrelation function.	05	
Q 5	**Attempt any 2 out of 3 parts**			
	a	i) What is a Poisson Process? ii) Let {N(t),t∈[0,∞)} be a Poisson process with rate λ=0.5. a. Find the probability of no arrivals in (3,5]. b. Find the probability that there is exactly one arrival in each of the following intervals: (0,1], (1,2], (2,3], and (3,4].	05	
		What do you understand by i An independent increment process, ii Stationary increment process iii Ergodic random process Explain with the help of suitable examples.	05	C05
		A 2-state homogeneous DTMC X_n is described by the state transition matrix $\mathbf{P} = \begin{bmatrix} 1-p & p \\ q & 1-q \end{bmatrix}$. i Find the PMF of the DTMC at time'3'. If the initial PMF vector is $[p_0\ p_1]$. ii Given that the DTMC is in state '0' at time '5', what is the probability that the random process shall be in state '1' at time '8'.	05	